A wider conditional support strip on the left probability panels

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This local proof was developed with AI assistance. Its arithmetic checks and its mathematical hypotheses are recorded separately. It is not a proof of the full sharp iid Berry–Esseen conjecture.

1 A wider strip on the remaining probability panels

The incoming strip can be extended to the left probability range without a new Fourier computation. The change is small: retain the negative conditional amplitude $3/5$, and allow positive amplitude $7/10$. The following statement is conditional on the same entry and contact hypotheses as the previously proved strip.

Proposition 1. *Let $n \ge 42$ and let (X, H) be an attained fixed-target witness with nonnegative excess at C_{E} and sharp predecessor $C_{n-1} \le C_{\mathrm{E}}$. Retain the selected v7 coarse entry, its actual central gap $(-A_-, A_+)$, and the direct one-pass geometry bounds of v10. For every retained closed probability panel $[i/100, (i+1)/100]$, $25 \le i \le 39$, put*

$$p = \mathbb{P}(X > 0), \quad q = 1 - p, \quad \ell = \frac{\mathbb{E}X_+}{pq}, \quad Y = X/\ell + p = B + U, \qquad B = \mathbf{1}_{\{X>0\}}.$$

Then the two conditional noises satisfy

$$-\frac{3}{5} < U_0 \le a_0, \qquad -a_1 \le U_1 < \frac{7}{10}, \qquad a_0, a_1 \ge 0, \quad a_0 + a_1 = d < \frac{17}{250}.$$

In particular $|U_j| < 7/10$ on the full left interval $[1/4, 2/5]$. The direct geometric exclusion already gives $p > 1/4$ for actual witnesses; retaining the closed endpoint in the cover only enlarges the tested domain.

Proof. Write $[l, u]$ for a panel and e, r, b_-, b_+ for its v10 lower bound on ℓ, upper bound on ρ, and lower and upper bounds on $\beta = \mathbb{E}[X|X|]$. For the negative ray use $b = -b_-$, and for the positive ray use $b = b_+$. The retained all-threshold heat lemma gives, for $t > \sqrt{3}$,

$$H_n(H, \pm t) \le h(t), \qquad |\partial_t H_n(H, \pm t)| < g(t),$$

where, with $S = 81/80$ and $\eta = 8/875$,

$$h(t) = S^3 \left\{ \frac{2}{5} \left(\frac{t^3 - 3t}{6} + \frac{|t^4 - 6t^2 + 3|}{72} \right) + \frac{t^5}{4100} + \frac{3t}{820} + \frac{17(3t^2 - 2)}{78720} \right\},$$

$$g(t) = \frac{2}{5} \left(\frac{t^2}{2} + \frac{1}{2} + \eta \right).$$

The contact upper estimate is $H_n(H, \pm t) + C_{\mathrm{E}} Q(t)$, with

$$Q(t) = -t^3 + \frac{3}{2} r t^2 + 3bt + \eta r.$$

These statements retain the actual index, nonnegative excess and sharp predecessor. In particular, they are not consequences of moment bounds alone. They hold for every raw threshold H.

Let $c_0 = 819/2000 < C_{\mathrm{E}}$ and $L = g + c_0 Q'$. At each proposed boundary the finite acceptance requires the following eight positive quantities:

$$t^2 - 3, \quad -h(t) - c_0 Q(t), \quad -Q(t), \quad -Q'(t), \quad -Q''(t), \quad -L(t), \quad -L'(t), \quad -L''(t).$$

Here $Q''' = -6$ and $L'' = 2/5 - 6c_0 = -2057/1000$. Thus Q, Q', Q'' remain negative along the entire outward ray; so do L, L'. The actual contact upper function has derivative strictly less than $L < 0$, because $Q' < 0$ and $C_{\mathrm{E}} > c_0$. Its boundary value is negative because $Q < 0$ and $h + c_0 Q < 0$. This excludes the entire ray, including its boundary. We differentiate the actual heat function using its derivative estimate, not the absolute-value expression in h.

Take

$$t_- = e(l + 3/5), \qquad t_+ = e(1 - u + 7/10).$$

The exact rational check passes all eight tests on both rays of all fifteen closed panels. The weakest negative boundary margin is greater than $3/10$, and the weakest positive boundary margin is greater than $27/100$. The input is precisely the source-bound v10 export; no update is fed back into that export. Shared panel endpoints occur in both neighbors, so the tests leave no uncovered seam.

It follows that $-t_- < X < t_+$ on the support. Since $\ell \geq e$ and $l \leq p \leq u$, division by the actual positive ℓ gives

$$-3/5 < Y < 1 + 7/10.$$

The actual inner gap gives

$$a_0 = p - A_-/\ell, \qquad a_1 = q - A_+/\ell, \qquad a_0 + a_1 = 1 - (A_- + A_+)/\ell = d < 17/250.$$

On the negative sign $U_0 = Y \leq a_0$, while on the positive sign $U_1 = Y - 1 \geq -a_1$. This proves the asymmetric bounds and hence the common amplitude claim. Conditional centering is unchanged. □

What the finite calculation does and does not say. The bounded exploration considered $0.65, 0.70, 0.75, 0.80, 0.85$ and stopped after the first common success, 0.70. It made only thirty positive-ray evaluations. At 0.65 the boundary test fails on panels 29–33 while the continuation tests pass. The first failure, on $[.29, .30]$, has boundary upper estimate strictly between .010460 and .010461. This is failure of that sufficient estimate, not a counterexample to the narrower strip. The acceptance calculation independently reconstructs the expressions at the selected boundaries; its 31 ray evaluations include this negative control. No Fourier cells, moment contraction rounds or support-generation campaign are run.

2 The resulting count domain

For the common amplitude $A = 7/10$ and displacement ceiling $d_0 = 17/250$,

$$A + 2d_0 = \frac{209}{250} < 1, \qquad l_0 = \frac{A - 1 - 2d_0}{2} = -\frac{109}{500}, \qquad h_0 = l_0 + 1 = \frac{391}{500}.$$

The crossing separation is $\zeta = 1 - d_0 = 233/250$, and $-\zeta < l_0 < h_0 < \zeta$. Every $x = np + H/\ell$ has a shifted anchor $x = k + u$ with $u \in [l_0, h_0]$. The gap from the amplitude constraints in the shifted-anchor construction is $(1 - A - 2d_0)/2 = 41/500 > 0$. Thus the pathwise crossing and curvature lemmas apply on this longer phase interval.

These anchors are legal. The retained raw-threshold bound is $|H| \le 1.83605680$, and every left-panel scale bound exceeds $49/25$. Hence $|H/\ell| < 1$. For $n \ge 42$ and $p \in [1/4, 2/5]$,

$$np - 1 - h_0 \ge 42/4 - 1 - h_0 > 0, \qquad n(1-p) - 1 + l_0 \ge 42(3/5) - 1 + l_0 > 0.$$

Consequently $1 \le k \le n-1$. This is an exact domain check, not a bound for the final count comparison. In particular, the variance and normalization constants proved earlier only for $p \ge .4$ cannot be reused without a new calculation.

The noise has the same scaling throughout:

$$s = \mathbb{E}U^2 = \theta/\ell^2, \qquad \lambda = ns = n\theta/\ell^2, \qquad \ell = (pq+s)^{-1/2}.$$

Together with the existing $3/5$ strips on panels 40–52, the result supplies a common $7/10$ conditional amplitude on every retained panel 25–52. The signed rays are bounds for one physical law. This statement does not identify a law with its reflection or replace the original closed-event reflection and attainment argument.

2.1 A Bernoulli gap where the external theorem applies

Schulz's strict finite-n Bernoulli theorem supplies $C_B < C_{\mathrm{E}}$ for every $0 < p < 1$. Its sharper bound applies on $[1/3, 2/3]$ and gives

$$\rho_B(C_{\mathrm{E}} - C_B) > \frac{2C_{\mathrm{E}}(r - p_*)^2}{\sqrt{pq}}, \qquad r = \min(p,q), \quad p_* = (4 - \sqrt{10})/2.$$

On the left subinterval $p \in [1/3, 2/5]$, both factors $(p_* - p)^2$ and $(p(1-p))^{-1/2}$ decrease. The minimum is therefore at $p = 2/5$. Using $C_{\mathrm{E}} > 819/2000$, $\sqrt{10} < 3.162277661$ and $\sqrt{6/25} < 49/100$, exact rational arithmetic gives the simple uniform reserve

$$\rho_B(C_{\mathrm{E}} - C_B) > \frac{59}{100000}.$$

Below $p = 1/3$ we retain only the global strict Bernoulli bound, with zero quantitative reserve. Panel 33 must be split at $1/3$ before using the sharper formula. Neither the sharper formula nor this positive uniform reserve is asserted below that endpoint.

2.2 A repair term that changes on the left

The favorable inequality $\rho \ge \rho_B$ used in the earlier small-noise argument does not hold on this entire new domain. For an exact example, take $p = 13/50$, $q = 37/50$, $U_1 = 0$ and let U_0 take $\pm\varepsilon$ with equal conditional probabilities, where $\varepsilon = 1/1000$. Put $v = pq$ and $s = q\varepsilon^2$. The centered raw law has nodes and masses

$$(-p - \varepsilon, q/2), \qquad (-p + \varepsilon, q/2), \qquad (q, p),$$

variance $V = v + s$ and third absolute moment $M_3 = v(1-2v) + 3ps$. Exact rational comparison gives

$$\frac{M_3^2}{V^3} < \frac{(1-2v)^2}{v},$$

so the standardized law satisfies $\rho < \rho_B$. Its inward displacement is only ε, and at $n = 42$ its noise is $\lambda = 42s = .00003108 < 1/500$. The law satisfies the new amplitude and inward bounds. No contact premise is claimed for this example. It refutes discarding the adverse moment repair for arbitrary laws in the enlarged count domain; it does not refute a count comparison.

Remaining obligations. The wider strip removes the missing support premise on the left panels. It does not prove their count inequality. A new argument must control the own count, incoming counts, normal and moment repairs on the wider probability and phase ranges. The selected coarse entry, contact and sharp predecessor remain ancestors of this strip. A finite base is still needed for the first-failure argument; no result here proves it.

Source locations. The retained heat/ray lemma is in `manuscript/arxiv-v1/sections/06-fixed-resources.tex`, lines 206–248, with its heat-expansion proof in Section 7 of that manuscript. The new one-pass input is `mathematics/simplification-v10/2026-10-28/geometry.tex`, its panel export being identified by hash in the accompanying checker. The shifted-anchor identity and curvature lemma are in the same directory's `counts.tex`. The external Bernoulli input is Jona Schulz, *The optimal Berry–Esseen constant in the binomial case*, dissertation (2016), Theorem 1, printed p. 15, German National Library copy.